

Quiz 10

● Graded

Student

Samyak Sanghvi

Total Points

4 / 4 pts

Question 1

Q1

2 / 2 pts

+ 0 pts No/Wrong answer

EITHER

✓ + 1 pt Watershed segmentation

✓ + 0.5 pts on gradient magnitude

✓ + 0.5 pts with known pixels as markers

OR

+ 1 pt graph cut

+ 1 pt with region costs only on known pixels

+ 1 pt Used edge detection (not guaranteed to give a closed curve enclosing a region)

+ 0 pts No/Wrong answer

a)

✓ + 0.25 pts large

✓ + 0.25 pts because gradients in all directions

b)

✓ + 0.25 pts large

✓ + 0.25 pts because both horizontal and vertical gradients

c)

✓ + 0.25 pts small

✓ + 0.25 pts because only horizontal gradients

d)

✓ + 0.25 pts large

✓ + 0.25 pts because gradients in all directions

Name: Sanyal

COL7(6)83 Quiz 9

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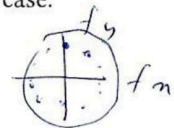
1. Suppose we want to segment an image in which the desired object and the background both have smoothly varying intensities spanning the full range $[0, 255]$, but there are large intensity gradients along the boundary between the two regions. Assume that the locations of one object pixel and one background pixel are known. Suggest a method for automatic segmentation, requiring as little parameter tuning or post-processing as possible.

Watershed Segmentation can be a good idea.

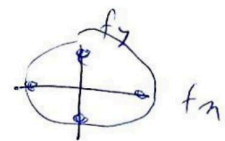
We can very easily localize ~~the region~~ and differentiate in between the regions inside and outside the boundary, and using the input pixels we can segment it.

2. Will the Harris-Stephens cornerness be large or small if the window is centered at (a) a bright spot on a dark background, (b) a checkerboard pattern, dark in two quadrants and light in the other two, (c) a dark line on a light background, (d) a randomly textured region with high contrast? Briefly explain why in each case.

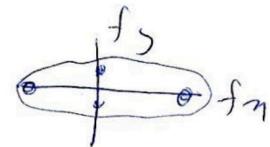
a) large $\Rightarrow$ have gradients in all dir



b) large $\Rightarrow$ have large gradient in $+f_x, +f_y$
 $-f_x, -f_y$ dir



c) small $\Rightarrow$ large $+f_x, -f_y$, little $+f_y, -f_x$



d) ~~small~~ large $\Rightarrow$ large variation in gradients in all dir

